Machine Learning Techniques for Image Forensics in Adversarial Setting (Ph.D. presentation)

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Machine Learning Techniques for Image Forensics in Adversarial Setting

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2, April 2020

Outline

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- Introduction
- Contribution

Part II: Overview

- Prior art on ML-based Image Forensics
- Adversarial Image Forensics

Part III: Original Contribution of the Thesis (SELECTED WORKS)

- Improving the Security of Image Manipulation Detection Through One-and-a-half-class Multiple Classification.
- On the Transferability of Adversarial Examples Against CNN-Based Image Forensics.
- Effectiveness of Random Deep Feature Selection for Securing Image Manipulation Detectors Against Adversarial Examples.

Introduction

- Multimedia Forensics gathers information on the **history** of multimedia documents.

From Model-based to Data-driven

- A statistical characterization and modeling for complex forensic tasks is often not available.

- Forensic researchers have resorted to ML techniques.

- **Disabiling ML-based forensic analysis** turns out to be an **easy task**!

- Overcome the security limits and design systems **thought to work** in the adversarial setting is a necessity.

Contribution of Thesis

- **We developed ML techniques for Image Forensics in adversarial setting (secure ML)**
  - **focus on image manipulation detection (and binary classification in particular).**
    - **Adversary-aware systems**
    - **Intrinsically more secure detectors [Focus]**

Part II
Prior Art on ML-Based Image forensics

Prior Art on ML-Based Image Forensics

Most common ML techniques used in image forensics.

- SVM-based image forensics
  - Many ML-based Image Forensic methods rely on SVM classification.
  - [few examples]: A-JPEG detection based on DCT coefficent or Histogram of low frequency DCT coeffiecient [1] ….
- CNN-based forensics [recently]
  - DL techniques and CNNs in particular are also used extensively for Steganalysis and Multimedia Forensics (MF).
  - [few examples]: Binary and multi-class CNN for detecting several manipulation operations [2] ….

[1] T. Pevny and J. Fridrich, "Detection of double-compression in JPEG images for applications in steganography," Trans. Info. For. Sec., vol. 3, no. 2, pp. 247-258, Jun. 2008.
[2] B. Bayar and M. C. Stamm, "A deep learning approach to universal image manipulation detection using a new convolutional layer," in Proceedings of the 4th ACM Workshop on Information Hiding and Multimedia Security, ser. IH&MMSec '16. New York, NY, USA: ACM, 2016, pp. 5-10.

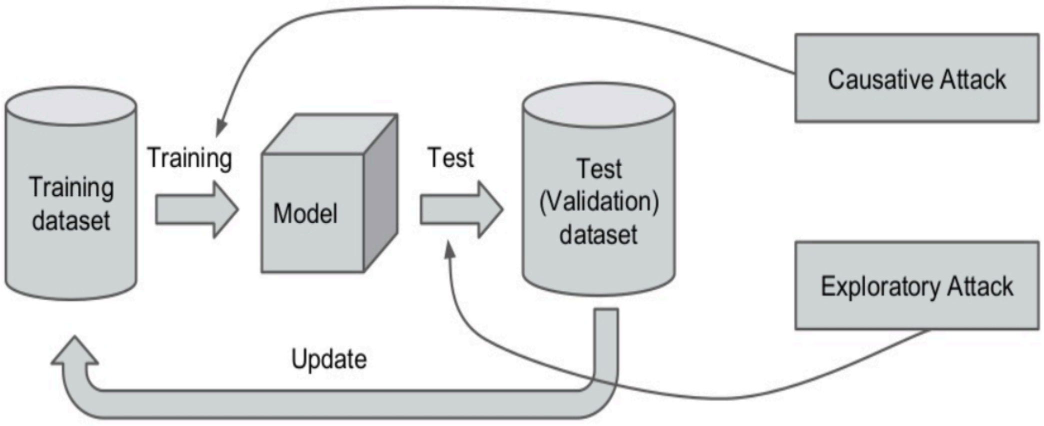
Adversarial Image Forensics

- What is **Counter-Forensics (CF)?**
  - Referred as **anti-forensics**.
  - Tools developed to **DISABLE** image forensic tools.
  - Many CF methods have been proposed, first against model-based tools then also against SVM-based and DL-based forensic tools.

Adversarial Image Forensics

➢ **Attacks can be classified based on several properties [1]**

- **Influence**
  - *Causative*
  - *Exploratory*



[1] L. Huang, A. D. Jpseph, B. Nelson, B. I. Rubinstein, and J. Tygar, "Adversarial machine learning", in Proceedings of the 4th ACM workshop in Security and artificial intelligence. ACM, 2011, pp. 43-58.

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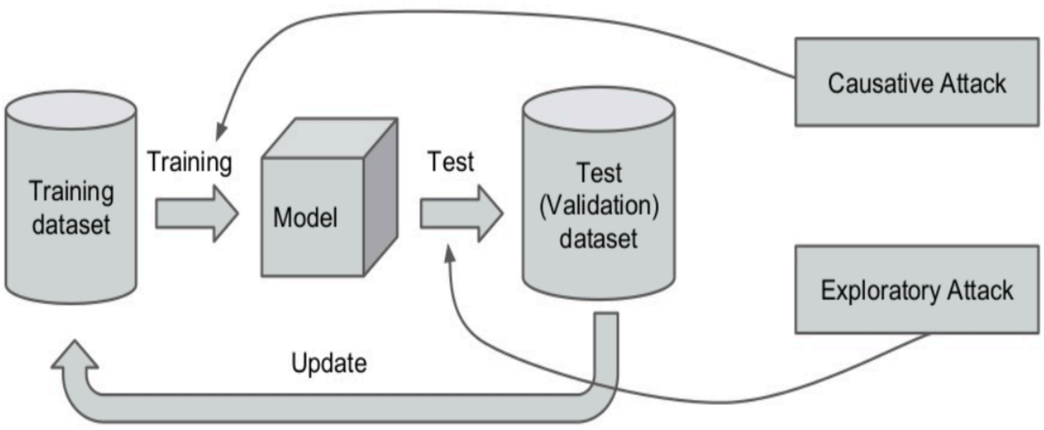

  - **Specificity**
    - *Targeted: attack focuses on the deception of a specific algorithm (classier).*
    - *Indiscriminative: when the attack is targeted to a class of algorithms (rather than a specific algorithm).*

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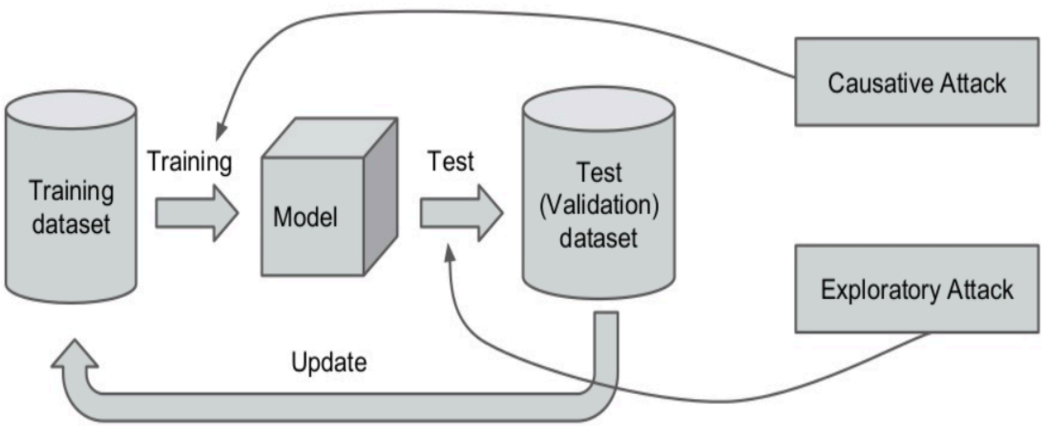

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  - *Targeted: attack focuses on the deception of a specific algorithm (classier).*
  - *Indiscriminative: when the attack is targeted to a class of algorithms (rather than a specific algorithm).*

- **Security violation**
  - *Integrity: False negative error.*
  - *Availability: both a false negative and a false positive error.*

[1] L. Huang, A. D. Jpseph, B. Nelson, B. I. Rubinstein, and J. Tygar, "Adversarial machine learning", in Proceedings of the 4th ACM workshop in Security and artificial intelligence. ACM, 2011, pp. 43-58.

Adversarial Image Forensics

Counter-forensic attack models (threat model)

➢ The Adversarial Model can be defined by specifying the following:

- **Attacker's goal:** it specifies the kind of **security violation**, hence the kind of error, the attacker aims at.
  - CF attacks are usually **integrity violation** attacks or **evasion** attacks.
- **Attacker's knowledge:** the attack can be **Perfect Knowledge (PK)** or **Limited Knowledge (LK).**
  - **PK**: the attacker has complete information about the forensic algorithm.
  - **LK**: attacker knows only some details about the forensic algorithm: e.g., he knows the parameters of the algorithm **and** does not know the training data.
- **Attacker's capability**: it applies mostly to ML, it refers to the control of the attacker over the training and/or testing data.

Adversarial Image Forensics

Attacks to deep learning-based image forensics (FOCUS ON)

- Recently, CF attacks against Deep Learning (DL) models have also been developed.

- A key advantage of CNNs is the ability to learn forensic features directly from the images.

- An intelligent attacker can use this property to his advantage and run powerful CF attacks, namely, **adversarial examples** **[1]**.

[1] C. Szegedy, W. Zaremba, I. Sutskever, J. Bruna, D. Erhan, I. Goodfellow, and R. Fergus, “Intriguing properties of neural networks," arXiv preprint arXiv:1312.6199 , 2013.

Adversarial Image Forensics

Attacks to deep learning-based image forensics (FOCUS ON)

- The vulnerability of DL to adversarial examples has recently been studied in forensics [1].
  - An attacker can easily create adversarial images by introducing a perturbation (high PSNR).
  - Many attacks have been proposed to fool **CNN-based detectors for image forensics** (camera model identification and manipulation detection).

[1] F. Marra, D. Gragnaniello, and L. Verdoliva, "On the vulnerability of deep learning to adversarial attacks for camera model identication," Signal Processing: Image Communication , vol. 65, pp. 240-248, July, 2018.

Adversarial Image Forensics

Anti-Counter Forensics (anti-CF)

We classify the anti-CF methods according to the **specificity** of the analyst's goal [1].

- **Adversary-aware systems**
  - The analyst develops a new algorithm to reveal the attack by looking the traces left by CF methods.
  - The analyst tries to **exit** the PK scenario or disinform the attacker.

- **Intrinsically more secure detectors**
  - The analyst looks for a system which is more difficult to attack **even** in the PK case.

[1] M. Barni, M. C. Stamm, and B. Tondi, "Adversarial multimedia forensics: Overview and challenges ahead," in 2018 26th European Signal Processing Conference (EUSIPCO). IEEE, 2018, pp. 962-966.

Part III

Original Contribution of the Thesis

Original Contribution of The Thesis

- **ML techniques for Image Forensics in adversarial setting (manipulation detection and binary classification)**
- Developed methods (classified based on the specificity of the analyst's goal)

  - **Adversary-aware training**
    - Most Powerful Attack (MPA)-aware Double JPEG detector (robust against a certain class of attacks) – SVM-based [1].
    - Detection of general (global&local) contrast adjustment robust to JPEG compression - via JPEG-aware training [2-3].
      - SVM-based and CNN-based solutions [**best one**]

[1] M. Barni, E. Nowroozi, and B. Tondi, "Higher-Order, Adversary-Aware, Double JPEG-Detection via Selected Training on Attacked Samples", In *25° European Signal Processing Conference (EUSIPCO),* Kos, Greece, August 2017.

[2] M. Barni, E. Nowroozi, and B. Tondi, "Detection of Adaptive Histogram equalization Robust Against JPEG Compression", In *International Workshop on Biometrics and Forensics (IWBF)*, Sassari, Italy, June 2018.

[3] M. Barni, A. Constanzo, E. Nowroozi, and B.Tondi, "CNN-based detection of generic contrast adjustement with JPEG post-processing", In *25° International Conference on Image Processing (ICIP),* Athens, Greece, October 2018.

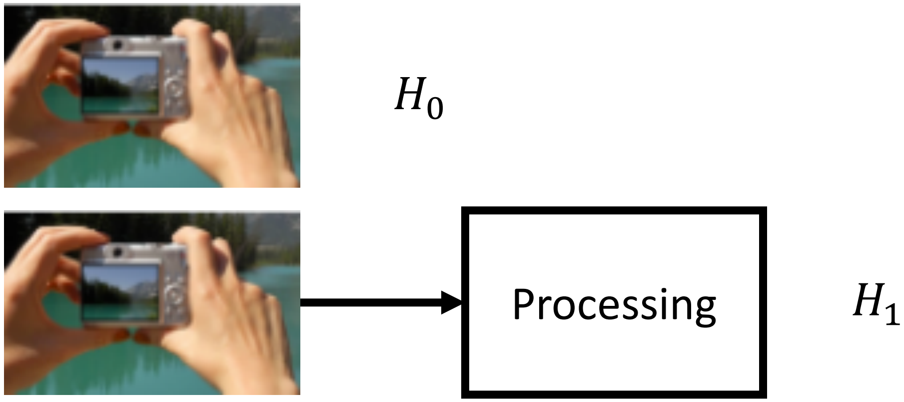
Original Contribution of The Thesis

- **ML techniques for Image Forensics in adversarial setting (manipulation detection and binary classification)**
- Developed methods (classified based on the specificity of the analyst's goal)

  - **Intrinsically more secure classifiers (FOCUS ON)**
    - Multiple-classifier architecture as a countermeasure against PK attacks (SVM-based).
    - Use of private information to secure ML forensic tools (SVMs and CNNs)

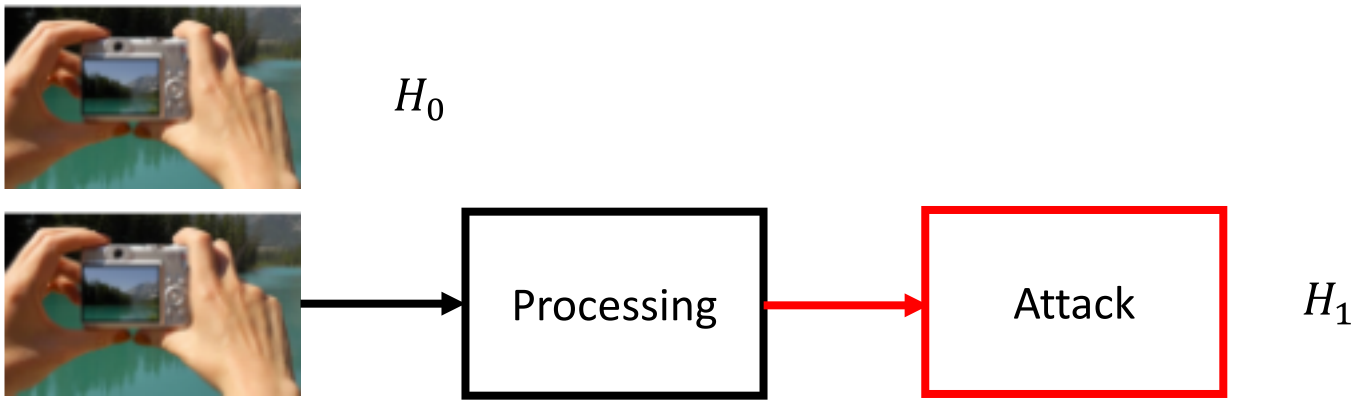
Improving the Security of Image Manipulation Detection Through One-and-a-half-class Multiple Classification

General Setup



- We focused on the case of binary classification.
- We consider $\boldsymbol{H_0}$ for Pristine and $\boldsymbol{H_1}$ for Manipulated/Processed images

Adversarial Setup



- The goal of the attacker is to induce a **missed detection error** **(integrity violation attack)**
- We denote with P_{md} the probability of a missed detection error ($P(H_0|H_1)$).
- We denote with P_{fa} the false alarm probability ($P(H_1|H_0)$).

Proposed System

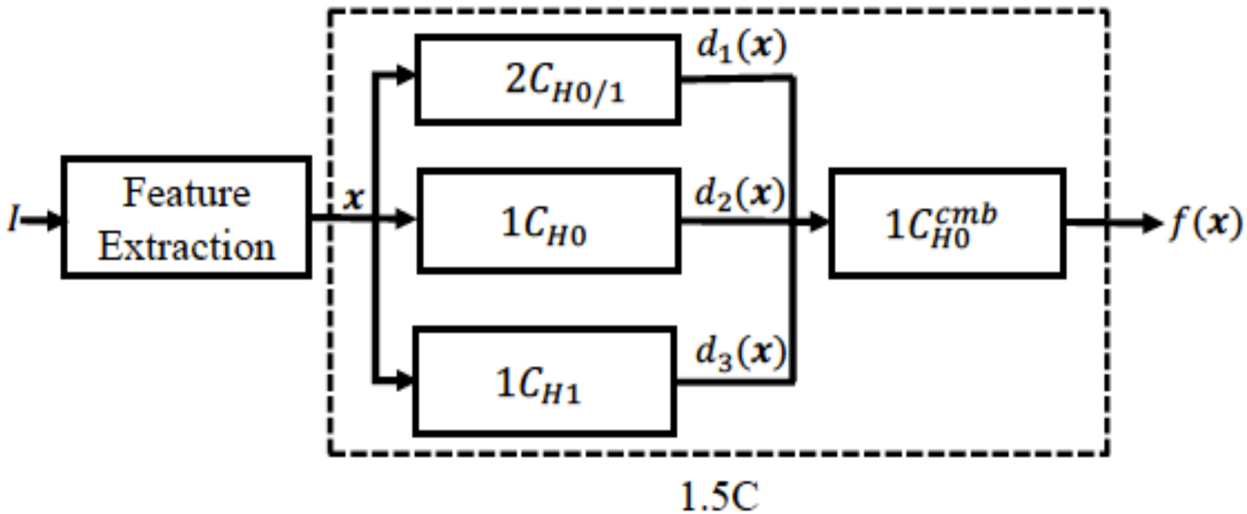
- To protect image manipulation detectors against PK attacks (getting intrinsically more secure detectors) by working on the architecture.

- We propose the use of a **one-and-a-half-class (1.5C) classifier.**

- Borrowed from the general literature of ML [1].

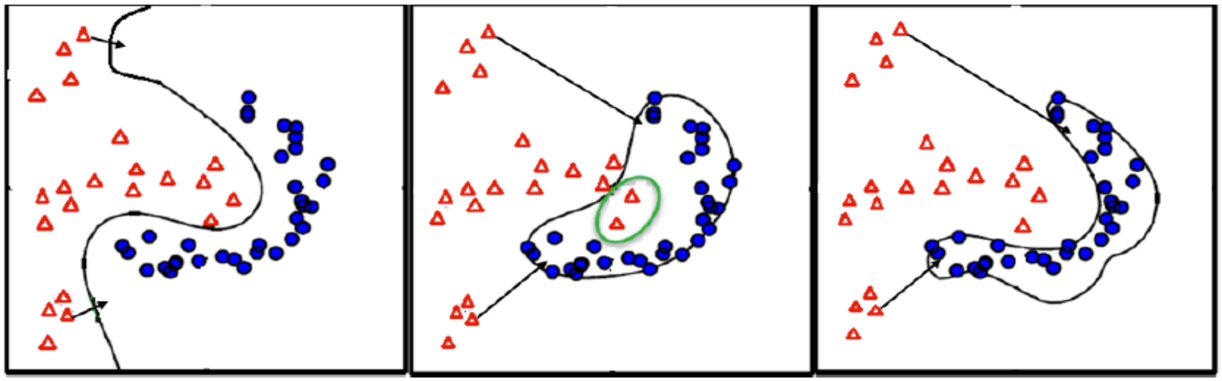
[1] B. Biggio, I. Corona, Z. -M. He, P. P. K. Chan, G. Giacinto, D. S. Yeung, and F. Roli, "One-and-a-half-class multiple classifier systems for secure learning against evasion attacks at test time", In Multiple Classifier Systems, Springer International Publishing, 2015, PP. 168-180.

Architecture 1.5C Classifier



- $\boldsymbol{I}$: image.
- $\boldsymbol{x}(\boldsymbol{I})$: a feature vector used to feed a multiple classifier.
- $\boldsymbol{2C_{H_0/1}}$: a 2C classifier, trained with examples H_0 and H_1.
- $\boldsymbol{1C_{H_0}}$: 1C classifier, trained with H_0 class.
- $\boldsymbol{1C_{H_1}}$: 1C classifier, trained with H_1 class.
- $\boldsymbol{1C_{H_0}^{cmb}}$: combination classifier.
- $\boldsymbol{d_1(x)}$, $\boldsymbol{d_2(x)}$, and $\boldsymbol{d_3(x)}$**:** the output of the $\boldsymbol{2C_{H_0/1}}$, $\boldsymbol{1C_{H_0}}$, and $\boldsymbol{1C_{H_1}}$.
- $\boldsymbol{f(x)}$: the decision function of the downstream 1C combination classifier trained on H_0.

Rational Behind The 1.5C Architecure



$2C$ $\quad$ $1C$ $\quad$ $1.5C$

- $\mathbf{2}\boldsymbol{C}$: achieves **high accuracy in the absence of attacks**.
- $\mathbf{1}\boldsymbol{C}$: **intrinsically more robust against attack.**
- $\mathbf{1.5}\boldsymbol{C}$: the goal is to **couple the advantages of $2C$ and $1C$**

Setup and Implementation

- **Evaluation Tasks:** three different detection tasks: **resizing**, **median filtering** and **histogram equalization**.

- **Implementation of 1.5C detector and choice of the feature set**

  - We implement all classifiers by means of **SVMs**.
  - Subtractive Pixel Adjacency Model (**SPAM**) feature set [1].

    (1) first-order residuals are computed in all directions.
    (2) Residual values are truncated (T=3).
    (3) Finally, second-order co-occurences are computed.

[1] T. Pevny, P. Bas, and J. Fridrich, “Steganalysis by subtractive pixel adjacency matrix," IEEE Transactions on Information Forensics and Security, vol. 5, no. 2, pp. 215-224, June 2010.

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- **Tests under attacks**

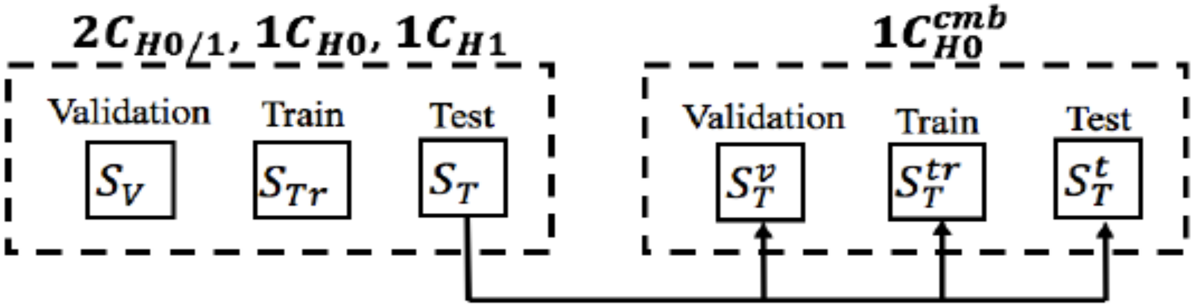
  - ad-hoc attack, gradient-inspired, pixel based attack to SVM detectors [2]

[1] T. Pevny, P. Bas, and J. Fridrich, “Steganalysis by subtractive pixel adjacency matrix," IEEE Transactions on Information Forensics and Security, vol. 5, no. 2, pp. 215-224, June 2010.
[2] Z. Chen, B. Tondi, X. Li, R. Ni, Y. Zhao, and M. Barni, “A gradient-based pixel-domain attack against SVM detection of global image manipulations,” in 2017 IEEE Workshop on Information Forensics and Security (WIFS) , Dec 2017, pp. 1-6.

Training The 1.5C Classifiers

- **Choice of the hyper-parameters in intermediate classifiers** for 2C $(\boldsymbol{\gamma}, \boldsymbol{C})$ and 1C $(\boldsymbol{\gamma}\ , \boldsymbol{v} = \mathbf{1}/\boldsymbol{C})$. Determined during validation phase.
  - $\boldsymbol{C}$: tradeoff between the margin of the separating hyperplane.
  - $\boldsymbol{\gamma}$: determines the width of the kernel and how far the influence of a training samples reaches.
  - $\boldsymbol{v}$: determines the margin of the decision region.
- **Training Methodology**
  - **2C**: we trained the system for every choice of $(\boldsymbol{C}, \boldsymbol{\gamma})$ and choose the pair with the best accuracy, which minimizes -> $\boldsymbol{P_e = 0.5P_{fa} + 0.5P_{md}}$
  - **1C**: exhaustive search applied on $\boldsymbol{\gamma}$ and $\boldsymbol{v}$ to find the best accuracy.
    - To avoid missed detection events, different weights for the two error probability terms need to be considered -> $\boldsymbol{P_e = \alpha P_{fa} + \beta P_{md}}, \quad \boldsymbol{\alpha < \beta}$

Dataset Creation



$$\{\boldsymbol{S_T^v} \cup \boldsymbol{S_T^{tr}} \cup \boldsymbol{S_T^t}\} \equiv \boldsymbol{S_T}$$

- RAISE-8K
- Total amount of images 7997 images split as follows
  - 1000 (S_v), 5000 (S_{Tr}), and 1997 (S_T)
  - Then the images in S_T: 300 (S_T^v), 700 (S_T^{tr}), and 997 (S_T^t)

Performance in The Absence of Attacks

- **AUC values of all classifiers for three manipulation detection tasks.**

	$2C_{H_{0/1}}$	$1C_{H_0}$	$1C_{H_1}$	$1C_{H_0}^{cmb}$
Resizing	**1**	0.96	0.96	**0.99**
Median filter	**1**	0.99	0.99	**0.99**
CLAHE	**1**	0.95	0.97	**0.99**

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- **Robustness of the classifiers [Resizing detection tasks - Similar performance on the other tasks]**

QF	$2C_{H_{0/1}}$	$1C_{H_0}$	$1C_{H_1}$	$1C_{H_0}^{cmb}$
85	0.90	0.71	0.83	0.88
90	0.94	0.75	0.87	0.93
95	0.98	0.82	0.90	0.97
98	0.99	0.87	0.91	0.99

(a)

Robustness of the classifiers in the presence of **JPEG compression (a)** for resizing detection task.

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(a)

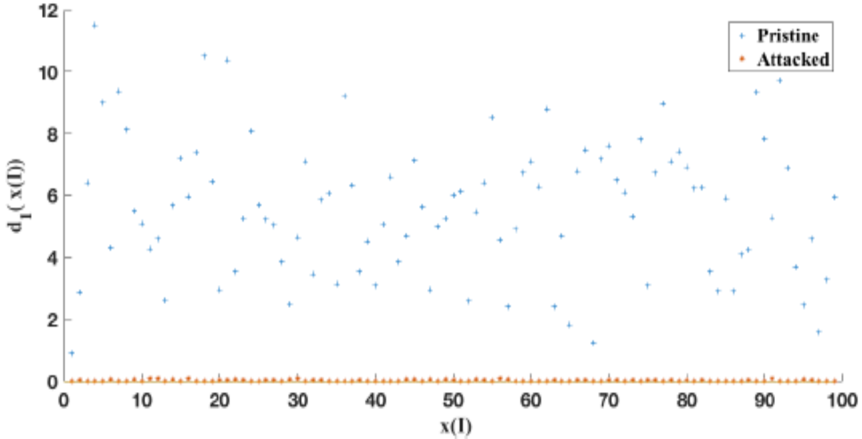
Variance	$2C_{H_{0/1}}$	$1C_{H_0}$	$1C_{H_1}$	$1C_{H_0}^{cmb}$
$5 \cdot 10^{-6}$	0.87	0.78	0.84	0.93
10^{-5}	0.84	0.79	0.85	0.92
$1.5 \cdot 10^{-5}$	0.79	0.80	0.87	0.89
$2 \cdot 10^{-5}$	0.74	0.82	0.89	0.83

(b)

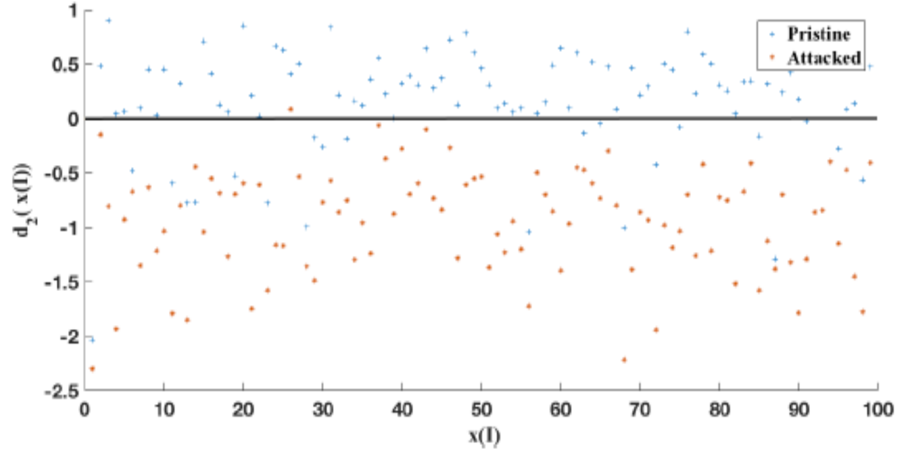
Robustness of the classifiers in the presence of **JPEG compression** (a), and under **noise addition** (b) for resizing detection task.

Performance Under of Attacks (1/3)

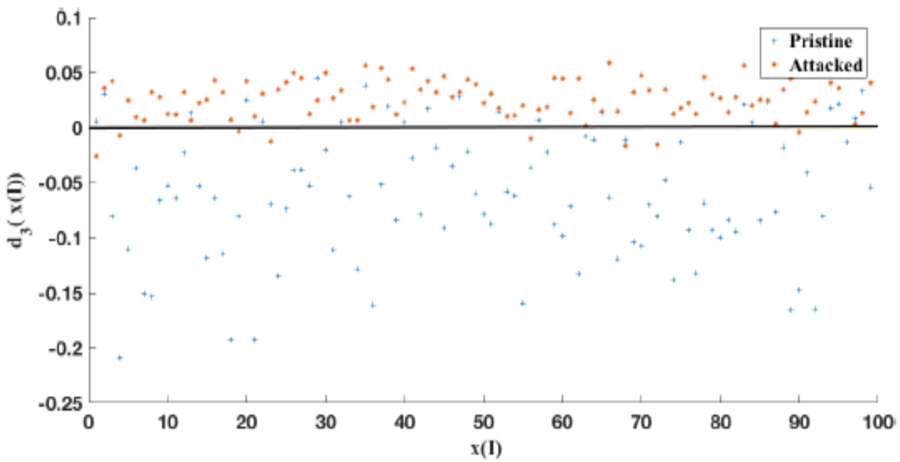
➢ **Attacks against $2C_{H_{0/1}}$ [only resizing task - Similar performance on the other tasks]**



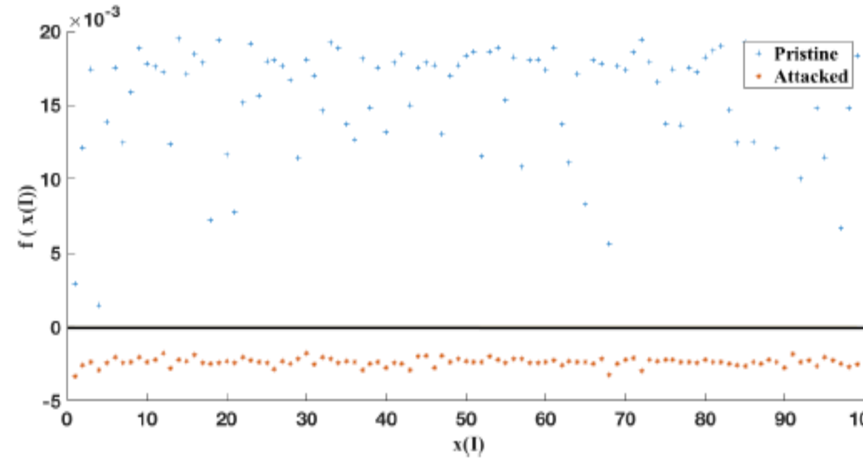
(a) $2C_{H_{0/1}}$



(b) $1C_{H_0}$



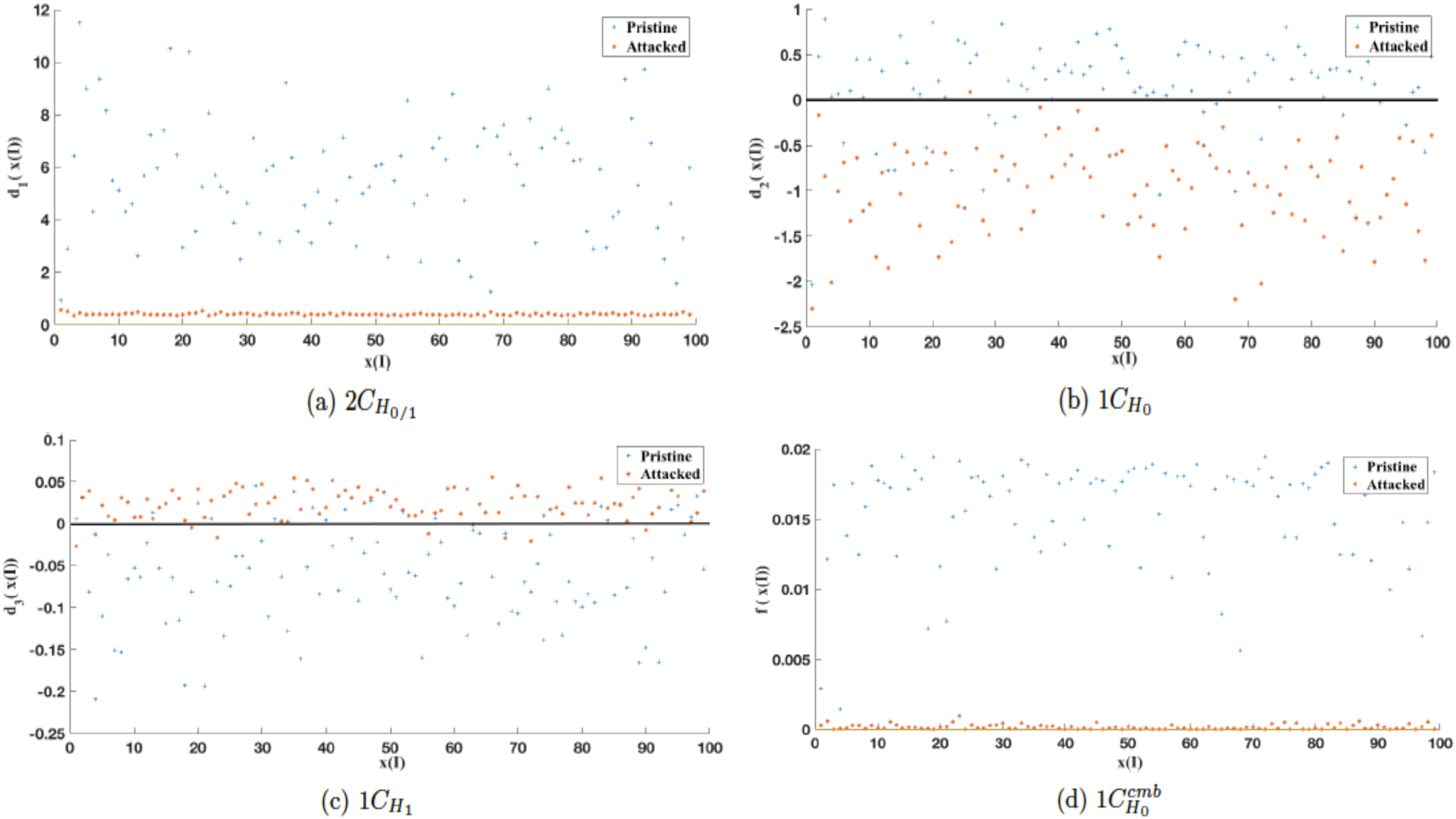
(c) $1C_{H_1}$



(d) $1C_{H_0}^{cmb}$

Performance Under of Attacks (2/3)

- Attacks against $1C_{H0}^{cmb}$ [only resizing task - Similar performance on the other tasks]



(a) $2C_{H_{0/1}}$

(b) $1C_{H_0}$

(c) $1C_{H_1}$

(d) $1C_{H_0}^{cmb}$

Performance Under Attacks (3/3)

- We assess the performance of $2C_{H_0/1}$ and the 1.5C classifiers against attacks,
  - The attack is PK (ad-hoc). **Attack Success Rate (ASR) = 100%**

- **ASR**

	$2C_{H_{0/1}}$	$1C_{H_0}$	$1C_{H_1}$	$1C_{H_0}^{cmb}$
Resizing	**100%**	1%	8%	**0%**
Median Filter	**100%**	4%	3%	**0%**
CLAHE	**100%**	20%	12%	**0%**

Percentage of misclassified attacked images. The attack is carried out against $2C_{H_0/1}$.

	$2C_{H_{0/1}}$	$1C_{H_0}$	$1C_{H_1}$	$1C_{H_0}^{cmb}$
Resizing	**100%**	1%	9%	**100%**
Median Filter	**100%**	20%	21%	**100%**
CLAHE	**100%**	23%	14%	**100%**

Percentage of misclassified attacked images. The attack is carried out against $1C_{H0}^{cmb}$.

Average MSE.

	Resizing	median Filter	CLAHE
Attack against $2C_{H_{0/1}}$	0.10	0.22	0.27
Attack against $1C_{H_0}^{cmb}$	0.17	0.60	0.43

Average of percentage of pixels modified by the attack.

	Resizing	median Filter	CLAHE
Attack against $2C_{H_{0/1}}$	9.5%	15.1%	12.3%
Attack against $1C_{H_0}^{cmb}$	13.4%	25.1%	15.1%

➢ For the case of 1.5C, the attack must introduce **more distortion** to be successful

On the Transferability of Adversarial Examples against CNN-based Forensics

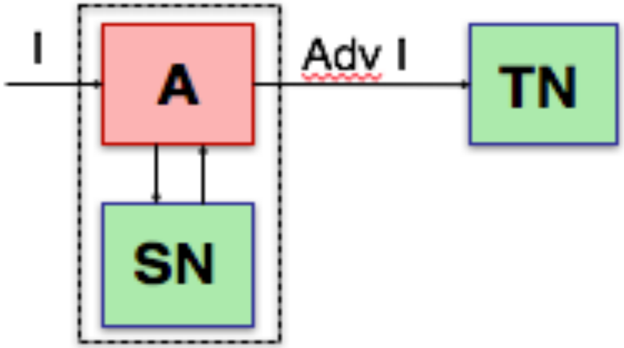
Introduction and Motivation

- Adversarial examples to CNNs are **transferable** (strong property, proven in ML literature) [1].
  - Attacks designed to fool a DL system can also fool others [**Blind** or **Transfer Attacks**]
- A problem in DL-based forensics (and DL for security): the tools work in an **adversarial envronment** !
  - **Does the transferability property also applies to CNNs models for forensics?**

[1] Papernot et al., "Transferability in machine learning: from phenomena to black-box attacks using adversarial samples", arXiv prepring arXiv: 1605.07277 (2016).

Goal

- Analyze the transferability of attacks against CNN-based image manipulation detectors,
  - Under different **sources of mismatch** between the network used to build the attack (**Source Network (SN)**) and the one the attack should be transferred to (**Target Network (TN)).**
  - Under different **types of attacks.**



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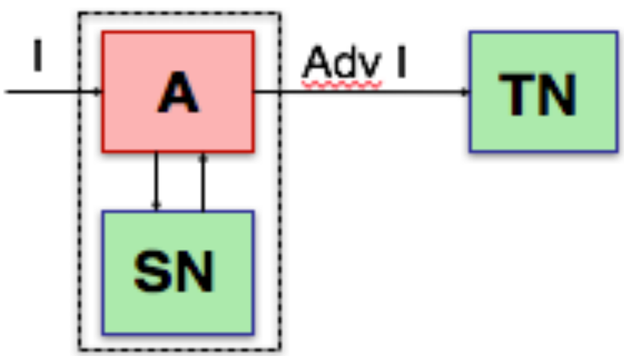

Adv [1]
- **I-FGSM**
- **JSMA**
- **L-BFGS**

[1] X. Yuan, P. He, Q. Zhu and X. Li, "Adversarial Examples: Attacks and Defenses for Deep learning," arXiv: 1712.07107, 2018.

Source of Mismatch

- **Cross-training**
  - The mismatch is in the dataset but networks match [partial match of SN and TN]
- **Cross-model**
  - The mismatch is in the Networks but training datasets match [partial match of SN and TN]
- **Cross-model-and training**
  - Mismatch networks trained on different datasets [complete mismatch between SN and TN]

Experimental Setup

- **CNN Architecture:** BSnet [1] (shallow network - **three covolutional layers**), and GCnet [2] (deep network – **nine convolutional layers**).

- **Processing task:** resizing (0.8) and median filtering (window size 5×5)

- **Dataset:** RAISE and Vision dataset (input patch size is 128×128)

    - E.g. N_{BS}^{R} **(res)** = BSnet trained on RAISE for resizing detection task.
    - The accuracy of SN and TN is above **97%** in all the cases.

- **Attack parameters,**

    - IFGSM: **S** = 10 (default) and $\boldsymbol{\varepsilon}$ = 0.01 and 0.001 -> Max Avg PSNR is 40db.
    - JSMA: **T** = 7 and $\boldsymbol{\theta}$ = 0.1 and 0.01 -> Max per-pixel distortion is below 70.
    - BFGS: $\boldsymbol{\varepsilon}$ = 0.1 (default parameter) -> Max Avg PSNR is 50db.

[1] B. Bayar and M. C. Stamm, "A deep learning approach to universal image manipulation detection using a new convolutional layer," in ACM Workshop on Info. Hiding & Multimedia Security, 2016, pp. 5–10.
[2] M. Barni, A. Costanzo, E. Nowroozi, B. Tondi. "Cnn-based detection of generic contrast adjustment with JPEG post-processing", ICIP2018, Athene, Greece.

Cross-Training Results

- **SN:** BSnet trained on RAISE (N_{BS}^{R}).
- **TN:** BSnet trained on VISION (N_{BS}^{V}) and viceversa.
- **ASR**: Attack Success Rate

SN	TN	Attack type	L1 dist	max. dist	ASR (SN)	ASR (TN)
N_{BS}^{R}(res)	N_{BS}^{V}(res)	I-FGSM, $\varepsilon_s = 0.01$	2.53	2.55	**1.000**	**0.692**
N_{BS}^{R}(res)	N_{BS}^{V}(res)	I-FGSM, $\varepsilon_s = 0.001$	0.26	0.27	1.000	0.0491
N_{BS}^{R}(res)	N_{BS}^{V}(res)	JSMA, $\theta = 0.1$	0.07	58.32	**1.000**	**0.782**
N_{BS}^{R}(res)	N_{BS}^{V}(res)	JSMA, $\theta = 0.01$	0.04	15.09	0.991	0.115
N_{BS}^{V}(res)	N_{BS}^{R}(res)	I-FGSM, $\varepsilon_s = 0.01$	2.53	2.55	1.000	0.002
N_{BS}^{V}(res)	N_{BS}^{R}(res)	I-FGSM, $\varepsilon_s = 0.001$	0.26	0.27	1.000	0.000
N_{BS}^{V}(res)	N_{BS}^{R}(res)	JSMA, $\theta = 0.1$	0.01	69.42	0.989	0.000
N_{BS}^{V}(res)	N_{BS}^{R}(res)	JSMA, $\theta = 0.01$	0.01	17.06	0.979	0.000
N_{BS}^{R}(med)	N_{BS}^{V}(med)	I-FGSM, $\varepsilon_s = 0.01$	2.53	2.55	**1.000**	**0.845**
N_{BS}^{R}(med)	N_{BS}^{V}(med)	I-FGSM, $\varepsilon_s = 0.001$	0.26	0.27	1.000	0.045
N_{BS}^{R}(med)	N_{BS}^{V}(med)	JSMA, $\theta = 0.1$	0.03	38.11	1.000	0.012
N_{BS}^{R}(med)	N_{BS}^{V}(med)	JSMA, $\theta = 0.01$	0.02	14.05	0.984	0.002
N_{BS}^{V}(med)	N_{BS}^{R}(med)	I-FGSM, $\varepsilon_s = 0.01$	2.53	2.55	**1.000**	**0.941**
N_{BS}^{V}(med)	N_{BS}^{R}(med)	I-FGSM, $\varepsilon_s = 0.001$	0.25	0.25	1.000	0.077
N_{BS}^{V}(med)	N_{BS}^{R}(med)	JSMA, $\theta = 0.1$	0.03	32.09	1.000	0.010
N_{BS}^{V}(med)	N_{BS}^{R}(med)	JSMA, $\theta = 0.01$	0.01	14.08	0.988	0.008

- The transferability is not symmetric with respect to the dataset.

Cross-Training Results

- **SN:** BSnet trained on RAISE (N_{BS}^{R}).
- **TN:** BSnet trained on VISION (N_{BS}^{V}) and viceversa.
- **ASR**: Attack Success Rate

➢ **Resizing task [BFGS]**

SN	TN	Att type	PSNR	L1 dist	max dist	ASR SN	ASR TN	ASR TN (Int)
$N_{\mathrm{BS}}^{\mathrm{R}}(\mathrm{res})$	$N_{\mathrm{BS}}^{\mathrm{V}}(\mathrm{res})$	BFGS $\varepsilon = 0.1$	57.81	0.25	2.13	1.000	0.261	0.169
$N_{\mathrm{BS}}^{\mathrm{V}}(\mathrm{res})$	$N_{\mathrm{BS}}^{\mathrm{R}}(\mathrm{res})$	BFGS $\varepsilon = 0.1$	57.62	0.25	2.26	1.000	0.002	0.002

Cross-Training Results

- **SN:** BSnet trained on RAISE (N_{BS}^{R}).
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$N_{\mathrm{BS}}^{\mathrm{R}}(\mathrm{res})$	$N_{\mathrm{BS}}^{\mathrm{V}}(\mathrm{res})$	BFGS $\varepsilon = 0.1$	57.81	0.25	2.13	1.000	0.261	0.169
$N_{\mathrm{BS}}^{\mathrm{V}}(\mathrm{res})$	$N_{\mathrm{BS}}^{\mathrm{R}}(\mathrm{res})$	BFGS $\varepsilon = 0.1$	57.62	0.25	2.26	1.000	0.002	0.002

Median-filtering task [BFGS]

SN	TN	Att type	PSNR	L1 dist	max dist	ASR SN	ASR TN	ASR TN (Int)
$N_{\mathrm{BS}}^{\mathrm{R}}(\mathrm{med})$	$N_{\mathrm{BS}}^{\mathrm{V}}(\mathrm{med})$	BFGS $\varepsilon = 0.1$	56.55	0.28	2.53	1.000	0.276	0.246
$N_{\mathrm{BS}}^{\mathrm{V}}(\mathrm{med})$	$N_{\mathrm{BS}}^{\mathrm{R}}(\mathrm{med})$	BFGS $\varepsilon = 0.1$	56.66	0.27	2.54	1.000	0.300	0.296

Cross-Model Results

- **SN:** BSnet trained on RAISE (N_{BS}^{R}).
- **TN:** GCnet trained on RAISE (N_{GC}^{R}).
- **ASR**: Attack Success Rate

Cross-Model Results

- **SN:** BSnet trained on RAISE (N_{BS}^{R}).
- **TN:** GCnet trained on RAISE (N_{GC}^{R}).
- **ASR**: Attack Success Rate

SN	TN	Attack type	L1 dist	max. dist	ASR (SN)	ASR (TN)
$N_{\mathrm{BS}}^{\mathrm{R}}$(res)	$N_{\mathrm{GC}}^{\mathrm{R}}$(res)	I-FGSM, $\varepsilon_s = 0.01$	2.53	2.55	1.000	0.002
$N_{\mathrm{BS}}^{\mathrm{R}}$(res)	$N_{\mathrm{GC}}^{\mathrm{R}}$(res)	I-FGSM, $\varepsilon_s = 0.001$	0.31	0.33	1.000	0.002
$N_{\mathrm{BS}}^{\mathrm{R}}$(res)	$N_{\mathrm{GC}}^{\mathrm{R}}$(res)	JSMA, $\theta = 0.1$	0.07	57.88	1.000	0.016
$N_{\mathrm{BS}}^{\mathrm{R}}$(res)	$N_{\mathrm{GC}}^{\mathrm{R}}$(res)	JSMA, $\theta = 0.01$	0.04	15.14	0.992	0.006
$N_{\mathrm{BS}}^{\mathrm{R}}$(med)	$N_{\mathrm{GC}}^{\mathrm{R}}$(med)	I-FGSM, $\varepsilon_s = 0.01$	2.53	2.55	**1.000**	**0.825**
$N_{\mathrm{BS}}^{\mathrm{R}}$(med)	$N_{\mathrm{GC}}^{\mathrm{R}}$(med)	I-FGSM, $\varepsilon_s = 0.001$	0.26	0.27	1.000	0.181
$N_{\mathrm{BS}}^{\mathrm{R}}$(med)	$N_{\mathrm{GC}}^{\mathrm{R}}$(med)	JSMA, $\theta = 0.1$	0.03	38.11	1.000	0.010
$N_{\mathrm{BS}}^{\mathrm{R}}$(med)	$N_{\mathrm{GC}}^{\mathrm{R}}$(med)	JSMA, $\theta = 0.01$	0.02	14.05	0.984	0.016

- Low transferability in general (only I-FGSM with larger ε is effective, for median filtering)
- The degree of transferability depends on the detection task (lower transferability for resizing).

Cross-Model Results

- **SN:** BSnet trained on RAISE (N_{BS}^{R}).
- **TN:** GCnet trained on RAISE (N_{GC}^{R}).
- **ASR**: Attack Success Rate

SN	TN	Attack type	L1 dist	max. dist	ASR (SN)	ASR (TN)
$N_{\mathrm{BS}}^{\mathrm{R}}$(res)	$N_{\mathrm{GC}}^{\mathrm{R}}$(res)	I-FGSM, $\varepsilon_s = 0.01$	2.53	2.55	1.000	0.002
$N_{\mathrm{BS}}^{\mathrm{R}}$(res)	$N_{\mathrm{GC}}^{\mathrm{R}}$(res)	I-FGSM, $\varepsilon_s = 0.001$	0.31	0.33	1.000	0.002
$N_{\mathrm{BS}}^{\mathrm{R}}$(res)	$N_{\mathrm{GC}}^{\mathrm{R}}$(res)	JSMA, $\theta = 0.1$	0.07	57.88	1.000	0.016
$N_{\mathrm{BS}}^{\mathrm{R}}$(res)	$N_{\mathrm{GC}}^{\mathrm{R}}$(res)	JSMA, $\theta = 0.01$	0.04	15.14	0.992	0.006
$N_{\mathrm{BS}}^{\mathrm{R}}$(med)	$N_{\mathrm{GC}}^{\mathrm{R}}$(med)	I-FGSM, $\varepsilon_s = 0.01$	2.53	2.55	**1.000**	**0.825**
$N_{\mathrm{BS}}^{\mathrm{R}}$(med)	$N_{\mathrm{GC}}^{\mathrm{R}}$(med)	I-FGSM, $\varepsilon_s = 0.001$	0.26	0.27	1.000	0.181
$N_{\mathrm{BS}}^{\mathrm{R}}$(med)	$N_{\mathrm{GC}}^{\mathrm{R}}$(med)	JSMA, $\theta = 0.1$	0.03	38.11	1.000	0.010
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- Low transferability in general (only I-FGSM with larger ε is effective, for median filtering)
- The degree of transferability depends on the detection task (lower transferability for resizing).

Cross-Model Results

- **SN:** BSnet trained on RAISE (N_{BS}^{R}).
- **TN:** GCnet trained on RAISE (N_{GC}^{R}).
- **ASR**: Attack Success Rate

➢ **Resizing model [BFGS]**

SN	TN	Att type	PSNR	L1 dist	max dist	ASR SN	ASR TN	ASR TN (Int)
$N_{\mathrm{BS}}^{\mathrm{R}}(\mathrm{res})$	$N_{\mathrm{GC}}^{\mathrm{R}}(\mathrm{res})$	BFGS $\varepsilon = 0.1$	57.80	0.25	2.13	1.000	0.061	0.063

Cross-Model Results

- **SN:** BSnet trained on RAISE (N_{BS}^{R}).
- **TN:** GCnet trained on RAISE (N_{GC}^{R}).
- **ASR**: Attack Success Rate

➢ Resizing model [BFGS]

SN	TN	Att type	PSNR	L1 dist	max dist	ASR SN	ASR TN	ASR TN (Int)
$N_{\mathrm{BS}}^{\mathrm{R}}(\mathrm{res})$	$N_{\mathrm{GC}}^{\mathrm{R}}(\mathrm{res})$	BFGS $\varepsilon = 0.1$	57.80	0.25	2.13	1.000	0.061	0.063

➢ Median-filtering task [BFGS]

SN	TN	Att type	PSNR	L1 dist	max dist	ASR SN	ASR TN	ASR TN (Int)
$N_{\mathrm{BS}}^{\mathrm{R}}(\mathrm{med})$	$N_{\mathrm{GC}}^{\mathrm{R}}(\mathrm{med})$	BFGS $\varepsilon = 0.1$	56.55	0.28	2.53	1.000	0.272	0.272

Cross-Model-and-Training Results

- **SN:** BSnet trained on VISION (N_{BS}^{V}).
- **TN:** GCnet trained on RAISE (N_{GC}^{R}).
- **ASR**: Attack Success Rate

Cross-Model-and-Training Results

- **SN:** BSnet trained on VISION (N_{BS}^{V}).
- **TN:** GCnet trained on RAISE (N_{GC}^{R}).
- **ASR**: Attack Success Rate

SN	TN	Attack type	L1 dist	max. dist	ASR (SN)	ASR (TN)
$N_{\mathrm{BS}}^{\mathrm{V}}$(res)	$N_{\mathrm{GC}}^{\mathrm{R}}$(res)	I-FGSM, $\varepsilon_s = 0.01$	2.53	2.55	1.000	0.004
$N_{\mathrm{BS}}^{\mathrm{V}}$(res)	$N_{\mathrm{GC}}^{\mathrm{R}}$(res)	I-FGSM, $\varepsilon_s = 0.001$	0.27	0.27	1.000	0.002
$N_{\mathrm{BS}}^{\mathrm{V}}$(res)	$N_{\mathrm{GC}}^{\mathrm{R}}$(res)	JSMA, $\theta = 0.1$	0.02	70.87	1.000	0.000
$N_{\mathrm{BS}}^{\mathrm{V}}$(res)	$N_{\mathrm{GC}}^{\mathrm{R}}$(res)	JSMA, $\theta = 0.01$	0.01	17.16	0.992	0.000
$N_{\mathrm{BS}}^{\mathrm{V}}$(med)	$N_{\mathrm{GC}}^{\mathrm{R}}$(med)	I-FGSM, $\varepsilon_s = 0.01$	2.53	2.55	**1.000**	**0.796**
$N_{\mathrm{BS}}^{\mathrm{V}}$(med)	$N_{\mathrm{GC}}^{\mathrm{R}}$(med)	I-FGSM, $\varepsilon_s = 0.001$	0.25	0.26	1.000	0.008
$N_{\mathrm{BS}}^{\mathrm{V}}$(med)	$N_{\mathrm{GC}}^{\mathrm{R}}$(med)	JSMA, $\theta = 0.1$	0.03	31.83	1.000	0.008
$N_{\mathrm{BS}}^{\mathrm{V}}$(med)	$N_{\mathrm{GC}}^{\mathrm{R}}$(med)	JSMA, $\theta = 0.01$	0.01	14.18	0.990	0.012

- The transferability of the attacks decreases further.

- Similar results are obtained by combining the architecure and dataset in the other way round.

Cross-Model-and-Training Results

- **SN:** BSnet trained on VISION (N_{BS}^{V}).
- **TN:** GCnet trained on RAISE (N_{GC}^{R}).
- **ASR**: Attack Success Rate

SN	TN	Attack type	L1 dist	max. dist	ASR (SN)	ASR (TN)
$N_{\mathrm{BS}}^{\mathrm{V}}$(res)	$N_{\mathrm{GC}}^{\mathrm{R}}$(res)	I-FGSM, $\varepsilon_s = 0.01$	2.53	2.55	1.000	0.004
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$N_{\mathrm{BS}}^{\mathrm{V}}$(med)	$N_{\mathrm{GC}}^{\mathrm{R}}$(med)	I-FGSM, $\varepsilon_s = 0.01$	2.53	2.55	**1.000**	**0.796**
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➢ **Resizing task [BFGS]**

SN	TN	Att type	PSNR	L1 dist	max dist	ASR SN	ASR TN	ASR TN (Int)
$N_{\mathrm{BS}}^{\mathrm{V}}$(res)	$N_{\mathrm{GC}}^{\mathrm{R}}$(res)	BFGS, $\varepsilon = 0.1$	57.62	0.25	2.26	1.000	0.0103	0.002

Cross-Model-and-Training Results

- **SN:** BSnet trained on VISION (N_{BS}^{V}).
- **TN:** GCnet trained on RAISE (N_{GC}^{R}).
- **ASR**: Attack Success Rate

Resizing task [BFGS]

SN	TN	Att type	PSNR	L1 dist	max dist	ASR SN	ASR TN	ASR TN (Int)
$N_{\mathrm{BS}}^{\mathrm{V}}$(res)	$N_{\mathrm{GC}}^{\mathrm{R}}$(res)	BFGS, $\varepsilon = 0.1$	57.62	0.25	2.26	1.000	0.0103	0.002

Median-filtering task [BFGS]

SN	TN	Att type	PSNR	L1 dist	max dist	ASR SN	ASR TN	ASR TN (Int)
$N_{\mathrm{BS}}^{\mathrm{V}}$(med)	$N_{\mathrm{GC}}^{\mathrm{R}}$(med)	BFGS, $\varepsilon = 0.1$	56.66	0.27	2.53	1.000	0.048	0.062

Lesson Learnt

- Poor transferabiliy of advesarial examples in Image Forensic applications.
- The degree of trasferability depends on several factors,
  - The **amount of mismatch**;
  - The **detection** and **classification task**;
  - The **attack method**,
    - JSMA is less transferable than I-FGSM.
    - BFGS is less transfereble than JSMA and I-FGSM.

Effectiveness of Random Deep Feature Selection for Securing Image Manipulation Detectors Against Adversarial Examples

Introduction and Motivation (1/2)

- Another possible approach to improve general robustness against CF attacks.

- Randomization strategies have been proposed to design **generally more secure** detectors in standard ML (for standard SVM-based classification) [1].

  - Randomization of the feature space according to a secret key has been proved to be effective to prevent the attack, both in theory and in practice.

  - The Random Feature Selection (RFS) scheme is validated by focusing on image manipulation detection and SVM classification.

- Random feature selection induces a **LK scenario** for the attack (due to the secrecy of the key).

[1] Z. Chen, B. Tondi, X. Li, R. Ni, Y. Zhao, and M. Barni, "Secure detection of image manipulation by means of random feature selection," IEEE Transactions on Information Forensics and Security, vol. 14, no. 9, pp. 2454-2469, Sep. 2019.

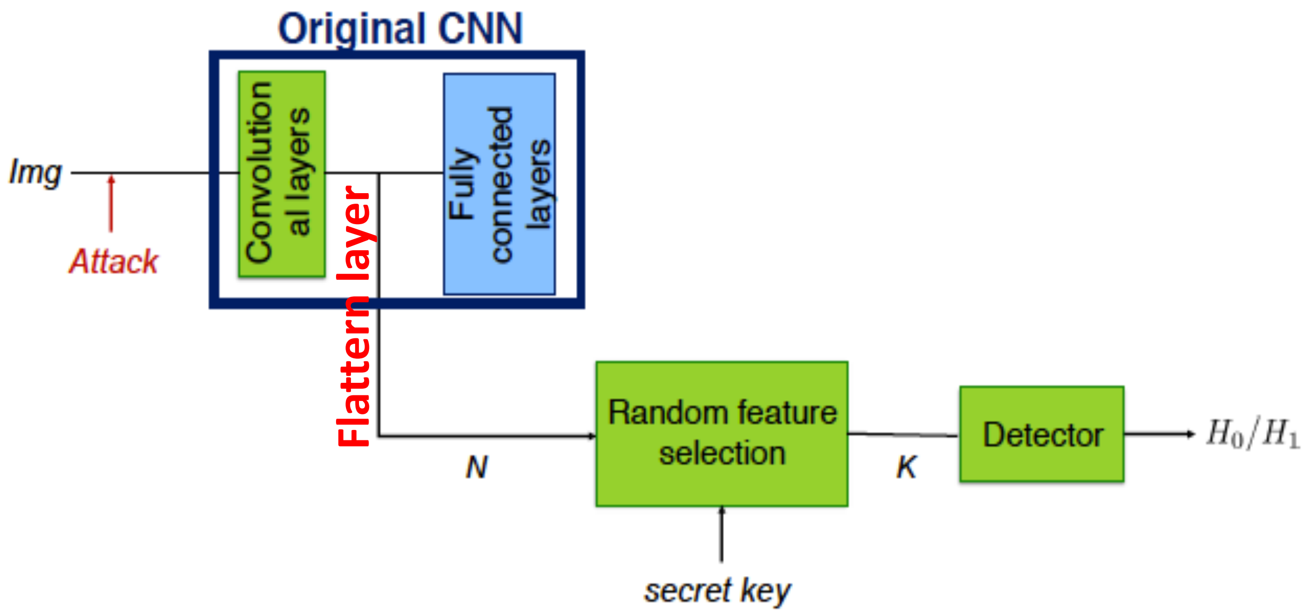
Introduction and Motivation (2/2)

- **Goal** : to extend the RFS in [1] to the case of CNN-based detection.
  - Features are extracted by a CNN and then randomized in the attempt to mitigate adversarial examples.
  - This scheme is referred to as **Random Deep Feature Selection** (RDFS).
- In DL literature, most of the methods based on randomization to combact adversarial examples focus on **test time randomization** [2].
  - Input layer of the classifier is randomized at test time.

[1] Z. Chen, B. Tondi, X. Li, R. Ni, Y. Zhao, and M. Barni, “Secure detection of image manipulation by means of random feature selection," IEEE Transactions on Information Forensics and Security, vol. 14, no. 9, pp. 2454-2469, Sep. 2019.
[2] C. Xie, J. Wang, Z. Zhang, Z. Ren, and A. Yuille, “Mitigating adversarial effectsthrough randomization," arXiv preprint arXiv:1711.01991, 2017.

Random Deep Feature Selection for Secure Image Classification

- We extended RFS method in [1] to the case of CNN-based forensic detectors.



- *K* features (*K* < *N*) are randomly selected among the *N* features

- The reduced set of features is used to train another detector (FC or SVM).

[1] Z. Chen, B. Tondi, X. Li, R. Ni, Y. Zhao, and M. Barni, "Secure detection of image manipulation by means of random feature selection," IEEE Transactions on Information Forensics and Security, vol. 14, no. 9, pp. 2454-2469, Sep. 2019.

RDFS-Based Detectors

RDFS detection based on FC network

- implemented by re-training the FC layers of original CNN on K features.
  - No mismatch in the architecture of the FC and the original classification part of the CNN (same FC layers)

RDFS detection based on SVM

- The amount of attack knowledge is less than in the FC case.
  - Attacker does not know the architecture, in addition to the training data (hp: the attack is carried out against the original CNN).

Experimental Setup

- **Network (for feature extraction):** BSnet [1] (shallow) and GCnet (deeper) [2].
  - **Input size:** 64×64
  - **Flatten Layer:** $N = 1729$ [1] and $N = 3200$ [2].

[1] B. Bayer and M. Stamm, "A deep learning approach to universal image manipulation detection using a new convolutional layer" in proceedings of the 4th ACM Workshop on Information Hiding and Multimedia Forensics, New York, NY, USA, 2016.
[2] M. Barni, M. Costanzo, E. Nowroozi and B. Tondi, "CNN-based detection of generic contrast adjustment with JPEG post-processing", in 25th IEEE International Conference on Image Processing (ICIP), pp. 3803-3807, Athens, Greece, 2018.

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Experimental Setup

- **Network (for feature extraction):** BSnet [1] (shallow) and GCnet (deeper) [2].
  - **Input size:** 64×64
  - **Flatten Layer:** N = 1729 [1] and N = 3200 [2].
- **Detection Tasks:** Resize, Median Filtering and Contrast Enhancement.
- **RDFS-based classifiers:** Fully Connected (FC) layer, and linear SVM (trained on a subset of images from the training set).
  - **Number of random features (K):** 10, 30, 50, 200, 400, 600, N = K.

[1] B. Bayer and M. Stamm, "A deep learning approach to universal image manipulation detection using a new convolutional layer" in proceedings of the 4th ACM Workshop on Information Hiding and Multimedia Forensics, New York, NY, USA, 2016.
[2] M. Barni, M. Costanzo, E. Nowroozi and B. Tondi, "CNN-based detection of generic contrast adjustment with JPEG post-processing", in 25th IEEE International Conference on Image Processing (ICIP), pp. 3803-3807, Athens, Greece, 2018.

Attack Types

- We considered three gradient-based iterative attacks,
  - **L-BFGS [1]:** optimum gradient-descent algorithm
  - **I-FGSM [2]:** sub-optimum attack
  - **PGD [3]:** with constrained maximum distortion (each iteration, the attack is projected into the bounded domain – allowed max distortion)

[1] C. Szegedy, W. Zaremba, I. Sutskever, J. Bruna, D. Erhan, I. Goodfellow, and R. Fergus, "Intriguing properties of neural networks", arXiv: 1312.6199, 2013.
[2] I. Goodfellow, J. Shlens, and C. Szegedy, "Explaining and harnessing adversarial examples", arXiv:1412.6572, 2014.
[3] A. Madry, A. Makelov, L. Schemidt, D. Tsipras, and A. Vladu, "Towards deep learning models resistant to adversarial attacks", in International Conference on Learning Representations, 2018.

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  - **I-FGSM [2]:** sub-optimum attack
  - **PGD [3]:** with constrained maximum distortion (each iteration, the attack is projected into the bounded domain – allowed max distortion)
- The attacks are run against the **original CNN** model in a **white-box** setting.
- Against the target CNNs, the **ASR** = **100%**.

[1] C. Szegedy, W. Zaremba, I. Sutskever, J. Bruna, D. Erhan, I. Goodfellow, and R. Fergus, "Intriguing properties of neural networks", arXiv: 1312.6199, 2013.
[2] I. Goodfellow, J. Shlens, and C. Szegedy, "Explaining and harnessing adversarial examples", arXiv:1412.6572, 2014.
[3] A. Madry, A. Makelov, L. Schemidt, D. Tsipras, and A. Vladu, "Towards deep learning models resistant to adversarial attacks", in International Conference on Learning Representations, 2018.

SVM (BSnet Features)

- **Average Accuracy** (averaged over 50 random selections K)

K	Resize				Median Filtering				CLAHE			
	No Attk	PGD	FGSM	BFGS	No Attk	PGD	FGSM	BFGS	No Attk	PGD	FGSM	BFGS
5	79.6	59.0	58.0	58.7	80.3	69.8	47.5	66.1	74.4	90.7	89.2	87.3
10	87.0	60.5	58.9	59.9	87.6	70.8	33.8	63.2	80.4	90.7	90.4	81.5
30	92.8	70.9	70.1	69.6	94.5	63.3	19.1	50.7	80.5	89.6	90.9	70.5
50	94.3	75.5	75.6	75.0	96.2	66.8	13.1	42.0	80.7	89.8	91.0	62.3
200	95.5	65.0	63.9	64.2	97.7	57.2	3.8	22.1	80.0	91.0	93.4	43.7
400	94.8	43.4	66.4	28.1	98.0	50.3	1.9	14.9	79.7	91.2	93.8	39.7
600	95.4	47.9	25.2	32.3	98.1	45.0	1.3	11.0	79.4	91.3	94.2	40.1
N	95.1	58.4	31.0	39.4	98.0	29.6	0.6	5.0	79.5	91.8	95.0	38.6

- When the attack works (against the full features detector), the randomization helps **(+30/40% gain in the Accuracy**), at the expense of a small reduced Accuracy without attacks (**-2/3%**)

FC (BSnet Features)

- **Average Accuracy** (averaged over 50 random selections K)

	Resize				Median Filtering				CLAHE			
K	No Attk	PGD	FGSM	BFGS	No Attk	PGD	FGSM	BFGS	No Attk	PGD	FGSM	BFGS
5	91.0	69.9	61.6	65.6	88.7	79.8	51.0	73.0	73.0	87.4	89.2	88.0
10	95.0	68.0	55.7	62.0	93.2	80.6	44.5	67.1	78.0	88.0	89.1	78.6
30	97.0	58.5	43.4	48.8	96.8	79.7	30.8	56.1	80.1	89.5	90.7	64.7
50	97.4	52.0	35.9	40.1	97.7	80.0	24.6	53.5	80.7	90.2	91.3	56.3
200	97.8	31.0	13.7	17.4	98.7	77.6	10.8	44.8	81.5	91.6	94.0	42.8
400	97.7	20.7	7.2	9.1	98.8	76.6	7.5	42.6	81.3	91.8	94.5	41.0
600	97.9	16.4	5.4	7.1	98.9	80.5	6.0	43.0	80.5	91.5	93.8	36.6
N	98.0	31.5	0.6	20.3	99.0	81.9	4.3	39.7	80.5	91.8	93.9	35.1

- Similar behaviour: the RDFS scheme is effective at the expense of a small reduction of Accuracy without attacks (**-2/4%**)

FC (BSnet Features)

- **Average Accuracy** (averaged over 50 random selections K)

	Resize				Median Filtering				CLAHE			
K	No Attk	PGD	FGSM	BFGS	No Attk	PGD	FGSM	BFGS	No Attk	PGD	FGSM	BFGS
5	91.0	69.9	61.6	65.6	88.7	79.8	51.0	73.0	73.0	87.4	89.2	88.0
10	95.0	68.0	55.7	62.0	93.2	80.6	44.5	67.1	78.0	88.0	89.1	78.6
30	97.0	58.5	43.4	48.8	96.8	79.7	30.8	56.1	80.1	89.5	90.7	64.7
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N	98.0	31.5	0.6	20.3	99.0	81.9	4.3	39.7	80.5	91.8	93.9	35.1

- Point: the Accuacy is already large also for K=N, i.e. the attacks fails against the **retrained** full features FC network.

Lesson(s) Learnt

- The **effectiveness** of RDFS depends on the detection task, attack type, and the network. In some cases, the **Accuracy** can be increased significantly, in other it can not.

- In some cases, **Randomization is unnecessary!**

  - The **mismatch in the architecture** (SVM, with K=N) decreases by itself the Accuracy.

  - **Re-training the same architecture** (FC, with K=N) on a different set (a subset) decreases by itself the Accuracy.

- *How can the attack transferability be improved?*

  - (necessary to properly assess defences)

Conclusion and Future Directions

Summary

- We investigated several **ML-based** techniques for image manipulation detection capable to work in **adversarial** setting.

  - Improving the robustness of ML-based image manipulation against laundering-type attacks via adversarial training.

  - Improving the security of ML-based image manipulation against a certain class of attacks via adversarial training.

  - Improving the security of ML-based image manipulation against PK (white-box) attacks through the use of 1.5C architecture.

  - Investigating the issue of attack transferability in DL-Image Forensics.

  - Exploiting feature randomization to mitigate the dangerousness of adversarial examples.

Open Issues and Future Research Direction

- Robustification of DL-based forensic methods to adversarial attacks

More focused perspective:

- The effectiveness of deep feature randomization should be assessed against stronger attacks
    - Increasing the strength of the attacks and then increasing the attack transferability turns out to be difficult task: towards **increased-confidence adversarial attacks** (**work in progress**).
    - Considering a scenario more favorable to the attacker and assess performance assuming that attacker is *aware* of the feature selection mechanism (only the secret unknown to him)

Future research direction:

- A security threat in DL that will deserve attention is the one posed by the use of GAN (**future reserach direction**).

List of Publications and Presentations (1/2)

- M. Barni, **E. Nowroozi**, and B. Tondi, "Higher-Order, Adversary-Aware, Double JPEG-Detection via Selected Training on Attacked Samples", In *25° European Signal Processing Conference (EUSIPCO),* Kos, Greece, August 2017. (**Presented by E. Nowroozi**)
- M. Barni, **E. Nowroozi**, and B. Tondi, "Detection of Adaptive Histogram equalization Robust Against JPEG Compression", In *International Workshop on Biometrics and Forensics (IWBF)*, Sassari, Italy, June 2018. (**Presented by E. Nowroozi**)
- M. Barni, A. Constanzo, **E. Nowroozi**, and B.Tondi, "CNN-based detection of generic contrast adjustement with JPEG post-processing", In *25° International Conference on Image Processing (ICIP),* Athens, Greece, October 2018. (**Presented by E. Nowroozi**)
- M. Barni, **E.Nowroozi**, and B. Tondi, "Improving the security of Image Manipulation Detection through One-and-a-half-class Multiple Classification", In *Multimedia Tools and Applications*, Springer, November 2019.

List of Publications and Presentations (2/2)

- M. Barni, K. Kallas, **E. Nowroozi**, and B. Tondi, “On the Transferability of Adversarial Examples Against CNN-Based Image Forensics”, In *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, Brighton, United Kingdom, May 2019.

- M. Barni, **E. Nowroozi**, B. Tondi, and B. Zhang, “Effectiveness of random deep feature selection for securing image manipulation detectors against adversarial examples”, Submitted to *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP),* Barcelona, Spain, arXiv: 1910.12392, October 2019.

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